Nicholas Saroiu

nsaroiu@hotmail.com — Cell: (206) 939-9398 — US & Canadian Citizen

Education

Honours Bachelor of Science — Computer Science Major, Mathematics Minor — University of Toronto Apr 2026
Relevant Coursework: Neural Networks and Deep Learning; Machine Learning; Computer Architecture; Operating Systems; Compilers and Interpreters; Parallel Programming.

Work Experience

Student Researcher — *Lee Language Lab* (PI: Prof. Annie Lee), Toronto, Ontario Sept 2025 – Apr 2026
- Implemented punctuation restoration as an unsupervised training objective for Romanian and Korean language models, testing whether it strengthens their capture of syntactic and semantic structure.
- Evaluated the objective on structure-sensitive tasks: part-of-speech tagging, named entity recognition, and chunking.

Jr. Software Engineer — *Dibbly Inc.*, Oakville, Ontario May 2025 – Aug 2025
- Built UI features for a large redesign of Dibbly Create, an AI-assisted manuscript editor for book writers, including task-settings popup windows, using React.js and Next.js.
- Introduced Pytest and Jest test infrastructure to the previously untested Create backend before moving to UI feature work.

Frontend Developer — *Roca Networks* (early-stage startup), Markham, Ontario June 2024 – Oct 2024
- Built a prototype real-time dashboard for residential battery systems with React.js and Chart.js, visualizing live performance and energy data.
- Designed a hierarchical access model scaling from a single property to admin, neighborhood, and multi-neighborhood views, intended to support town-scale deployments.

Projects

MTG Deck Completion — Live Demo
- Built an end-to-end recommender that completes 100-card Magic: The Gathering Commander decks from a partial list, trained on a self-collected, publicly released dataset of 189K real decklists.
- Engineered a resumable, rate-limited Moxfield API data pipeline with popularity-tiered sampling across 2,400 commanders.
- Designed a 32K-card tokenizer combining learned IDs, structured card features, and sentence-transformer embeddings of rules text.
- Trained DeepSets and Set Transformer models on masked set prediction, ensembled with a deck-pruning model; rebalancing the data raised long-tail MRR on unseen commanders 7×.
- Deployed a FastAPI + React/TypeScript app to Hugging Face Spaces; GitHub Actions tests each model release before deploy.

Causal Tracing for Transformer Interpretability
- Located where GPT-2 Small, Medium, Large, and XL store factual associations by corrupting inputs and restoring hidden states.
- Reproduced core experiments from the NeurIPS 2022 ROME paper "*Locating and Editing Factual Associations in GPT*" in PyTorch across all four model sizes.

Models to Predict Student Accuracy
- Compared K-Nearest Neighbors, Item-Response Theory, and autoencoder models for predicting student answer correctness in PyTorch and scikit-learn.
- Reached 70% validation accuracy by combining models in an ensemble and tuning with cross-validation.

Patient Portal Backend
- Built a Django REST Framework and PostgreSQL API for a dental patient-management system, designed for HIPAA privacy.
- Implemented JWT authentication and encryption of sensitive patient records.

TTC Tracker
- Built a full-stack Toronto transit tracker with a live React.js map and arrival predictions, following SOLID principles.
- Engineered a Java Spring Boot backend to ingest and process TTC GTFS real-time data feeds.

Skills

- **Languages:** Python, Java, C, C++, JavaScript, TypeScript, SQL, MIPS Assembly.
- **Frameworks/Tools:** PyTorch, scikit-learn, NumPy, Hugging Face, FastAPI, Django, React.js, Next.js, Spring Boot, PostgreSQL, Docker, GitHub Actions, SLURM, LLVM.
- **Concepts:** Deep Learning, Transformer Interpretability, Recommender Systems, NLP, Parallel Programming, REST API Design.